

■ 1.4.6 Simplifying Algebraic Expressions

There are many situations where you want to write a particular algebraic expression in the “simplest possible form”. Although it is difficult to know exactly what one means in all cases by the “simplest form”, a worthwhile practical procedure is to look at many different forms of an expression, and pick out the one that involves the smallest number of parts.

Simplify[expr] try to find the form of *expr* with the smallest number of parts, by applying a sequence of different algebraic transformations

Simplifying algebraic expressions.

Simplify writes $x^2 + 2x + 1$ in factored form.

```
In[1]:= Simplify[x^2 + 2x + 1]
Out[1]= (1 + x)^2
```

Simplify leaves $x^{10} - 1$ in expanded form, since for this expression, the factored form is larger.

```
In[2]:= Simplify[x^10 - 1]
Out[2]= -1 + x^10
```

You can often use **Simplify** to “clean up” complicated expressions that you get as the results of computations.

Here is the integral of $\frac{1}{x^4-1}$. Integrals are discussed in more detail in Section 1.5.

```
In[3]:= Integrate[1/(x^4-1), x]
Out[3]= 
$$\frac{-\text{ArcTan}[x]}{2} + \frac{\text{Log}[-1 + x]}{4} - \frac{\text{Log}[1 + x]}{4}$$

```

Differentiating the result from **Integrate** should give back your original expression. In this case, as is common, you get a more complicated version of the expression.

```
In[4]:= D[%, x]
Out[4]= 
$$\frac{1}{4(-1 + x)} - \frac{1}{4(1 + x)} - \frac{1}{2(1 + x^2)}$$

```

Simplify succeeds in getting back the original, more simple, form of the expression.

```
In[5]:= Simplify[%]
Out[5]= 
$$\frac{1}{-1 + x^4}$$

```

For many simple algebraic calculations, you may find it convenient to use **Simplify** quite routinely on your results.

In more complicated calculations, however, you often need to exercise more con-

control over the exact form of answer that you get. In addition, when your expressions are complicated, **Simplify** may spend a long time testing a large number of possible forms, in its attempt to find the simplest one.